

Arrhythmias — VF & Electrical Storm



1. Electrical storm = ≥ 3 in 24h

WHAT YOU SEE

Clock '24h' with ' ≥ 3 VT/VF' and 'MEDICAL EMERGENCY'

WHAT IT ENCODES

Electrical storm is defined as 3 or more separate episodes of sustained VT/VF (or 3 or more appropriate ICD shocks) within 24 hours — it is a true medical emergency with high short-term mortality. It reflects a hyperadrenergic, electrically unstable myocardium. Management is layered: defibrillate, suppress with antiarrhythmics (amiodarone) PLUS beta-blockade, deeply sedate, fix reversible triggers, and escalate to ablation/mechanical support if refractory.

BOARD TRAP

WRONG: defining electrical storm as one long VT episode — it requires 3 or more discrete sustained episodes (or ICD shocks) in 24 hours. A single sustained VT, however prolonged, is incessant VT, not storm.

2. Burning heart = VF

WHAT YOU SEE

Heart on fire labeled 'VF'

WHAT IT ENCODES

Ventricular fibrillation is chaotic, disorganized ventricular electrical activity with no organized QRS and no effective cardiac output, producing immediate pulseless cardiac arrest. The ECG shows irregular, fibrillatory waves of varying amplitude (coarse early, fine later). Survival depends on the time to defibrillation — every minute of delay drops survival ~7–10%.

BOARD TRAP

WRONG: attempting synchronized cardioversion for VF — there is no R wave to synchronize to, so the defibrillator may withhold the shock. VF is always treated with immediate UNSYNCHRONIZED defibrillation.

3. Defibrillation first

WHAT YOU SEE

Firefighters blasting the flames FIRST with their hoses before anything else — dousing the fire first = immediate unsynchronized defibrillation, which precedes any drug in VF/pulseless VT.

WHAT IT ENCODES

VF and pulseless VT are treated with immediate unsynchronized defibrillation (200 J biphasic) plus high-quality CPR — electrical therapy precedes drugs. In ACLS, after the first/second shock and CPR cycles, give epinephrine 1 mg IV every 3–5 min and an antiarrhythmic (amiodarone 300 mg then 150 mg, or lidocaine) for shock-refractory VF. Mnemonic: 'Shock, CPR, drugs' — in that order.

BOARD TRAP

WRONG: giving amiodarone or epinephrine before the first shock in VF arrest — defibrillation is the only intervention that restores a perfusing rhythm in VF and must come first. Drugs are adjuncts for shock-refractory VF, not the initial step.

4. Most common cause = acute MI

WHAT YOU SEE

'POLYMORPHIC VT DEGENERATING INTO VF DURING ACUTE MI' / 'MOST COMMON CAUSE'

WHAT IT ENCODES

Acute myocardial ischemia/infarction is the most common cause of VF — ischemic polymorphic VT during an acute MI degenerates into VF, the leading mechanism of pre-hospital sudden cardiac death. Definitive treatment is emergent revascularization plus beta-blockade. VF in the first 48 hours of an MI is from acute ischemia and does NOT by itself mandate an ICD if it is fully revascularized.

BOARD TRAP

WRONG: implanting an ICD for VF that occurred within 48 hours of an acute MI that was revascularized — this is reversible ischemic VF, not a primary arrhythmic substrate, so it does not meet secondary-prevention ICD criteria. Reassess EF ≥ 40 days post-MI for primary prevention.

5. Amiodarone in storm

WHAT YOU SEE

A drug bottle in the firefighters' hands as they keep dousing flames — the chemical retardant = IV amiodarone, the antiarrhythmic layered on after shocks to suppress refractory VF/storm.

WHAT IT ENCODES

IV amiodarone (300 mg bolus, then 150 mg repeats and a continuous infusion ~1 mg/min) is the antiarrhythmic of choice to suppress refractory VF and electrical storm after defibrillation. Procainamide or lidocaine are alternatives. Amiodarone works best when COMBINED with beta-blockade, which targets the catecholamine surge that amiodarone alone does not.

BOARD TRAP

WRONG: relying on amiodarone alone for electrical storm — the synergistic, often outcome-changing addition is sympathetic blockade (beta-blocker). Amiodarone monotherapy frequently fails in storm without concurrent beta-blockade and sedation.

6. Beta-blocker (sympathetic block)

WHAT YOU SEE

A firefighter physically blocking flames that keep re-igniting — blocking the re-flare = beta-blockade quenching the catecholamine surge that keeps re-lighting the storm.

WHAT IT ENCODES

Beta-blockade is a cornerstone of electrical storm: it blunts the catecholamine surge that drives recurrent VT/VF. Non-selective propranolol or IV esmolol infusion is preferred; in refractory storm, left stellate ganglion / cardiac sympathetic denervation is the next step. This is the single most under-used yet highest-yield therapy in storm.

BOARD TRAP

WRONG: avoiding beta-blockers in electrical storm out of fear of hypotension — sympathetic blockade is the key therapy that reduces recurrent shocks and mortality. Use IV esmolol (titratable) or propranolol; do not withhold beta-blockade in storm.

7. Sedation / intubation

WHAT YOU SEE

The crew sedating and managing the patient on the stretcher — calming the patient deeply = deep sedation/intubation that cuts the sympathetic drive fueling the storm (an active antiarrhythmic step).

WHAT IT ENCODES

Deep sedation (and intubation if needed) reduces endogenous sympathetic drive and pain/anxiety from repeated shocks, directly lowering the catecholamine load that fuels storm. Agents that also dampen sympathetic tone (e.g., propofol, dexmedetomidine) are favored. Sedation is an active antiarrhythmic measure in storm, not just comfort care.

BOARD TRAP

WRONG: treating sedation as purely supportive comfort in storm — reducing sympathetic drive through deep sedation is part of the antiarrhythmic strategy and is recommended alongside beta-blockade and amiodarone.

8. Ablation for refractory storm

WHAT YOU SEE

An 'ABLATION CATHETER' aimed like a torch at the fire's source — burning out the triggering focus/scar circuit = catheter ablation for storm refractory to drugs and beta-blockade.

WHAT IT ENCODES

Catheter ablation of the triggering PVC focus or the reentrant VT circuit/scar is used for electrical storm refractory to drugs and beta-blockade, and reduces recurrent shocks. In hemodynamically unsupportable storm, mechanical circulatory support (ECMO/Impella) may bridge to ablation. Ablation targets the substrate that drugs cannot quiet.

BOARD TRAP

WRONG: escalating antiarrhythmic drugs indefinitely in drug-refractory storm — guideline-favored next step after failed drugs/beta-blockade is catheter ablation, not stacking more antiarrhythmics.

9. ICD terminates, doesn't prevent

WHAT YOU SEE

An 'ICD ALARM SYSTEM' panel noting 'ICD = 1 shock = +risk' — the alarm box terminates each VT/VF episode but doesn't prevent recurrence; repeated shocks signal storm and worsen prognosis.

WHAT IT ENCODES

An ICD terminates VT/VF by delivering a shock but does NOT prevent the arrhythmia from recurring — it is a safety net, not a cure. Recurrent appropriate shocks define storm, cause myocardial injury and PTSD-like distress, and independently worsen prognosis. Adjunctive drugs/ablation are needed to reduce the burden of episodes.

BOARD TRAP

WRONG: assuming an ICD alone is sufficient management once placed — it does not reduce arrhythmia frequency, and recurrent shocks signal storm requiring antiarrhythmics, beta-blockade, and often ablation.

10. Reversible causes (ischemia, K/Mg)

WHAT YOU SEE

A 'CORRECT K+ / Mg2+' note with 'BB ↓ VF RISK' — the checklist to hunt and fix reversible storm drivers (ischemia, low K/Mg, proarrhythmic drugs) before escalating.

WHAT IT ENCODES

Always hunt for and treat reversible drivers of storm: acute ischemia (revascularize), electrolyte derangements (correct K+ to ~4.5–5.0 and replete Mg2+), QT-prolonging or proarrhythmic drugs, decompensated heart failure, and hyperthyroidism. Beta-blockade lowers VF risk across these settings. Reversible-cause correction can terminate storm by itself.

BOARD TRAP

WRONG: jumping to ablation or escalating antiarrhythmics before checking electrolytes and ischemia — uncorrected hypokalemia/hypomagnesemia or ongoing ischemia will defeat any antiarrhythmic. Fix reversible triggers first.

11. HCM — leading SCD in young

WHAT YOU SEE

A castle fort labelled 'HCM' with 'SCD <1%/yr' — the thick-walled fortress = the hypertrophied, fibrotic HCM myocardium, a leading cause of sudden death in the young.

WHAT IT ENCODES

Hypertrophic cardiomyopathy is a leading cause of sudden cardiac death in young people and athletes, from VT/VF on a hypertrophied, fibrotic, disarrayed myocardium (annual SCD risk often <1%/yr but concentrated in high-risk patients). ICD is indicated for high-risk features: prior cardiac arrest/sustained VT, unexplained syncope, massive LVH (≥30 mm), family history of SCD, or nonsustained VT/abnormal BP response.

BOARD TRAP

WRONG: clearing an HCM patient on EF alone — HCM patients usually have preserved/hyperdynamic EF yet remain at high SCD risk. Risk stratify by the specific HCM risk markers, not ejection fraction.

12. Lamin A/C — high VT/SCD

WHAT YOU SEE

A 'GENETIC MAP' panel reading 'LAMIN A/C / SCN5A / 30-40% of NICM' — the genetic chart flags LMNA cardiomyopathy as a malignant high-VT/SCD genotype warranting a low ICD threshold.

WHAT IT ENCODES

Lamin A/C (LMNA) cardiomyopathy is a malignant genetic dilated cardiomyopathy carrying a high risk of VT/SCD and progressive conduction disease (AV block) — it warrants a LOW ICD threshold even when EF is relatively preserved. Other arrhythmogenic mimics: SCN5A, PLN, desmoplakin. Suspect it in DCM with early conduction disease or family history of SCD.

BOARD TRAP

WRONG: applying the standard EF ≤35% ICD cutoff to LMNA cardiomyopathy — its high arrhythmic risk justifies ICD at higher EF and earlier than typical nonischemic cardiomyopathy. Genotype changes the ICD threshold.

13. ICD driving restriction

WHAT YOU SEE

A 'NEVER DRIVE' sign with '>220' near the ICD — the no-driving warning = the post-ICD/post-event driving restriction (permanent ban on commercial driving) patients must be counseled on.

WHAT IT ENCODES

After an ICD is placed (or after a sustained VA event), patients face temporary driving restrictions because of syncope/shock risk while driving. Typical guidance: after secondary-prevention ICD or an arrhythmic event, no private driving for ~3–6 months and a permanent ban on commercial driving; after primary-prevention ICD, ~1 week (post-implant). Counseling on driving is a commonly tested practical point.

BOARD TRAP

WRONG: telling an ICD patient they may resume commercial/heavy-vehicle driving — secondary-prevention ICD recipients are permanently disqualified from commercial driving, and private driving is restricted for a defined recovery period.

14. In-hospital recovery / monitoring

WHAT YOU SEE

A '48h' in-hospital recovery panel with a patient in a hospital gown on telemetry — the monitored 48-hour window = continuous ICU-level watch after resuscitated VF for early recurrence.

WHAT IT ENCODES

After resuscitated VF/storm, patients need continuous telemetry and ICU-level monitoring (the '48h' window) to catch early recurrence, manage post-arrest care, and stabilize the substrate (revascularization, electrolytes, beta-blockade). Targeted temperature management and neuroprognostication follow out-of-hospital arrest. Early recurrence is common, so monitored recovery is essential.

BOARD TRAP

WRONG: discharging or down-grading a resuscitated VF patient quickly because they look well — early arrhythmia recurrence and post-arrest complications require continuous monitored recovery and full reversible-cause workup before any de-escalation.
